BIOL-1 Exam 01

College of the Redwoods, Del Norte
Instructor: Dr. Daniel Friedman

Name: ____

Date: ____

Modules 01-06

Total Points: 50

Part A: Multiple Choice (30 points)

Select the best answer for each question. Each question is worth 1 point.

Module 01: The Science of Biology & Big Picture

1. Which of the following correctly orders the levels of biological organization from smallest to largest?
    - A) Cell → Molecule → Organ → Tissue
    - B) Atom → Molecule → Cell → Tissue → Organism
    - C) Tissue → Cell → Organism → Ecosystem
    - D) Population → Community → Organism → Cell
2. A scientist observes that plant growth increases with sunlight. She predicts that plants given 10 hours of light will grow taller than those given 5 hours. This prediction is a(n):
    - A) Conclusion

- B) Theory
- C) Hypothesis
- D) Control

3. Homeostasis is an organism's ability to:

    - A) Evolve over many generations
    - B) Convert solar energy into chemical energy
    - C) Maintain a stable internal environment despite external changes
    - D) Reproduce sexually

4. What is the fundamental unit of life?

    - A) The Atom
    - B) The Cell
    - C) The Molecule
    - D) The Organism

5. Which domain of life includes complex, membrane-bound cells such as plants, animals, and fungi?

    - A) Bacteria
    - B) Archaea
    - C) Eukarya
    - D) Viruses

Module 02: Chemistry of Life

1. The atomic number of an element is determined by its number of:

    - A) Protons
    - B) Neutrons
    - C) Electrons
    - D) Isotopes

2. When carbon and oxygen share electrons to form carbon dioxide, they form what type of bond?
    - A) Ionic bond
    - B) Covalent bond
    - C) Hydrogen bond
    - D) Peptide bond
3. Why does ice float on liquid water?
    - A) It is heavier than liquid water.
    - B) It is less dense than liquid water due to stable hydrogen bonds.
    - C) It is nonpolar and repels water.
    - D) It shrinks as it freezes.
4. A solution with a high concentration of H^+ ions and a pH of 3 is considered a(n):
    - A) Acid
    - B) Base
    - C) Neutral solution
    - D) Isotope
5. Which subatomic particles are found in the nucleus of an atom?
    - A) Electrons and protons
    - B) Protons and neutrons
    - C) Electrons and neutrons
    - D) Only protons

Module 03: Biological Molecules

1. Which of the following macromolecules is the primary source of quick cellular energy?
    - A) Proteins
    - B) Lipids
    - C) Carbohydrates
    - D) Nucleic Acids

2. The building blocks (monomers) of proteins are:
    - A) Monosaccharides
    - B) Fatty acids
    - C) Amino acids
    - D) Nucleotides
3. Which class of macromolecules includes fats, oils, and the main components of cell membranes?
    - A) Carbohydrates
    - B) Lipids
    - C) Nucleic Acids
    - D) Proteins
4. What occurs when a protein is "denatured"?
    - A) It replicates to form a new protein.
    - B) It binds to its substrate perfectly.
    - C) It loses its 3D shape and its function because of heat or extreme pH.
    - D) It changes into a carbohydrate.
5. DNA and RNA store and transmit genetic information. They belong to which category of macromolecules?
    - A) Proteins
    - B) Lipids
    - C) Nucleic Acids
    - D) Carbohydrates

Module 04: The Cell & Organelles

1. What distinguishing feature makes a eukaryotic cell different from a prokaryotic cell?
    - A) It has a cell membrane.
    - B) It has ribosomes.
    - C) It has genetic material (DNA).

- D) It has a membrane-bound nucleus and specialized organelles.

2. Which organelle is recognized as the "power plant" of the cell, where most ATP is produced?

    - A) Chloroplast
    - B) Mitochondrion
    - C) Ribosome
    - D) Golgi apparatus

3. Proteins are assembled on which cellular structure?

    - A) Lysosomes
    - B) Ribosomes
    - C) The smooth endoplasmic reticulum
    - D) Vacuoles

4. In plant cells, the rigid outer layer that provides shape and structural support is the:

    - A) Cell Wall
    - B) Cell Membrane
    - C) Cytoskeleton
    - D) Extracellular Matrix

5. Which organelle acts as the cell's "shipping center," packaging and distributing proteins?

    - A) Nucleus
    - B) Golgi apparatus
    - C) Mitochondrion
    - D) Chloroplast

Module 05: The Cell Membrane & Transport

1. The core structure of the cell membrane is composed of a:

    - A) Carbohydrate web
    - B) Rigid layer of proteins

- C) Phospholipid bilayer
- D) Single layer of nucleic acids

2. The movement of molecules from an area of high concentration to an area of low concentration is called:
    - A) Active transport
    - B) Diffusion
    - C) Endocytosis
    - D) Phagocytosis
3. Osmosis is a specific type of diffusion that involves the movement of:
    - A) Water molecules across a semi-permeable membrane
    - B) Large proteins out of the cell
    - C) Sodium ions entering the cell
    - D) Oxygen gases leaving the cell
4. If an animal cell is placed in a strongly hypertonic solution (like very salty water), what will happen?
    - A) The cell will swell and possibly burst.
    - B) The cell will stay the exact same size.
    - C) The cell will lose water and shrink (shrivel).
    - D) The cell will start undergoing photosynthesis.
5. Which transport process requires the cell to spend energy (ATP)?
    - A) Simple diffusion
    - B) Facilitated diffusion
    - C) Osmosis
    - D) Active transport

Module 06: Energy & Metabolism

1. What is the term for all of the chemical reactions that happen inside a living cell?
    - A) Photosynthesis
    - B) Thermodynamics
    - C) Cellular Respiration
    - D) Metabolism
2. What is an enzyme?
    - A) A protein that speeds up chemical reactions.
    - B) A carbohydrate used for structural support.
    - C) A lipid that stores energy.
    - D) A nucleic acid that carries genetic instructions.
3. The main purpose of cellular respiration is to break down glucose to generate:
    - A) Sunlight
    - B) Oxygen
    - C) Building blocks for proteins
    - D) ATP
4. Photosynthesis takes place inside which organelle?
    - A) Mitochondrion
    - B) Ribosome
    - C) Chloroplast
    - D) Nucleus
5. In aerobic respiration, what is the critical role of oxygen?
    - A) It provides the energy to build glucose.
    - B) It serves as the final electron acceptor in the electron transport chain.
    - C) It is a source of carbon to build proteins.
    - D) It breaks down the cell wall.

Part B: Fill in the Blank (11 points)

Complete each statement with the best biological term from the word bank below. Each question is worth 1 point.

Word Bank:

- Active
- Autotroph
- Catabolic
- Denaturation
- Endosymbiotic
- Fermentation
- Hypothesis
- Ionic
- Lysosomes
- Monosaccharides

- Mosaic

- A(n) _____________ is a proposed explanation for a scientific observation, which can be tested through experimentation.

- The type of bond where electrons are transferred (donated and accepted) between atoms, resulting in charged ions, is a(n) _____________ bond.

- The specific region on an enzyme where the substrate binds is known as the _____________ site.

- Complex carbohydrates like starch and glycogen are polymers made up of repeating simple sugar units called _____________.

- _____________ are membrane-bound "trash recycling" organelles that contain digestive juices to break down old cellular materials.

- The concept that mitochondria and chloroplasts originated as free-living bacteria that were engulfed by a larger cell is called the _____________ theory.

- In the fluid ____________ model of the cell membrane, proteins float around within a flexible lipid sea.
- When an enzyme operates perfectly due to an ideal environment, but then is heated up too much and changes shape, this shape-change process is called ____________.
- The breaking down of molecules to release energy is called a(n) ____________ reaction.
- ____________ is an anaerobic process (it occurs without oxygen) that allows cells to continue making a small amount of ATP and forms products like lactic acid or ethanol.
- An organism that can produce its own food using light energy is a(n) ____________ (or producer).

Part C: Free Response (9 points)

Choose THREE of the following five questions. Answer clearly in the space provided. Each response is worth 3 points.

1. **Water: The Molecule of Life.** Describe two unique properties of water (e.g., cohesion, high specific heat, universal solvent) and briefly explain why each property is essential for living organisms.
2. **The Flow of Genetic Information.** Trace the general path from DNA to a functional protein. Name the main "manager" organelle, the "factory worker" organelle that builds the protein, and the "shipping" organelle.
3. **Transport Options.** Contrast simple diffusion and active transport. In your answer, mention concentration gradients (high to low vs. low to high) and the use of ATP.
4. **Energy Cycles.** Briefly describe the relationship between Photosynthesis and Cellular Respiration. How do the waste products of one process serve as the essential ingredients for the other?

5. **Molecular Form and Function.** Choose any one of the major macromolecules (Carbohydrates, Lipids, Proteins, Nucleic Acids). Give an example of a specific molecule in that category, and explain how its physical shape/structure allows it to do its job.

Response 1

Question Number: _____

Response 2

Question Number: _____

Response 3

Question Number: _____

Feedback (Optional)

A. Is there anything else you would like to share regarding your exam preparation, or concepts you found particularly interesting/challenging?

B. What score (or percentage) do you estimate you earned on this exam? How confident are you in that estimate (1–10)?

End of Exam 01